Final — 100 MCQs

Internal Medicine (Cardiology) · Physical Therapy

01 Anatomy & Physiology of the Heart

10 Q

1. The heart weighs approximately:

a) About 2 kg
b) About 3 kg
c) **Less than a pound ✓**
d) About 5 pounds

2. The heart lies within the:

a) **Inferior mediastinum ✓**
b) Pleural cavity
c) Abdominal cavity
d) Superior mediastinum

3. The apex of the heart is directed toward the:

a) Right shoulder
b) Left shoulder
c) Right hip
d) **Left hip ✓**

4. The base of the heart points toward the right shoulder and lies beneath the:

a) 7th rib
b) Diaphragm
c) **2nd rib ✓**
d) 5th rib

5. The fibrous pericardium:

a) Forms the heart valves
b) **Protects the heart and anchors it to the diaphragm and sternum ✓**
c) Produces blood pressure
d) Pumps blood

6. The heart is approximately the size of a person's:

a) **Fist ✓**
b) Foot
c) Head
d) Forearm

7. The actual pumps of the heart (discharging chambers) are the:

a) Atria
b) Pericardium
c) Valves
d) **Ventricles ✓**

8. The septum that separates the two ventricles is the:

a) **Interventricular septum ✓**
b) Mediastinum
c) Pericardium
d) Interatrial septum

9. The heart valves are responsible for:

a) **Securing one-way blood flow ✓**
b) Producing blood cells
c) Gas exchange
d) Producing blood pressure

10. Which is NOT a function of the heart?

a) Producing blood pressure
b) Separating pulmonary and systemic circulations
c) **Producing red blood cells ✓**
d) Managing blood supply

02 Embryology & Congenital Heart Disease

18 Q

11. The first organ to form within the embryo is the:

a) Kidney
b) Brain
c) **Heart ✓**
d) Liver

12. Heart developmental abnormalities affect about …… of every 1000 births:

a) **8–10 ✓**
b) 100
c) 1–2
d) 50–60

13. Tetralogy of Fallot represents about …… of all congenital heart disease:

a) 1%
b) **10% ✓**
c) 90%
d) 50%

14. All are fetal shunts EXCEPT:

a) Foramen ovale
b) **Coronary sinus ✓**
c) Ductus venosus
d) Ductus arteriosus

15. In fetal circulation, heart pressures are:

a) Equal on both sides
b) **Higher on the right side than the left ✓**
c) Zero on the right side
d) Higher on the left side than the right

16. After birth, the fetal shunts close and remain as:

a) Heart valves
b) Coronary arteries
c) **Ligaments and fossae ✓**
d) Open channels

17. Congenital heart disease is an abnormality that is:

a) **Present at birth, even if discovered much later ✓**
b) Acquired after rheumatic fever
c) Caused only by infections
d) Always discovered at birth

18. Down's syndrome is associated with:

a) Aortic root abnormalities
b) **ASD and VSD ✓**
c) Mitral stenosis
d) Coarctation of aorta

19. Turner's syndrome is associated with:

a) PDA
b) **Coarctation of aorta and pulmonary stenosis ✓**
c) Tricuspid regurgitation
d) ASD and VSD

20. Marfan syndrome is associated with:

a) Tetralogy of Fallot
b) Pulmonary stenosis only
c) **Aortic root abnormalities ✓**
d) PDA

21. Which of the following is a cyanotic congenital heart disease?

a) VSD
b) ASD
c) **Tetralogy of Fallot ✓**
d) PDA

22. **In left-to-right shunts, all the shunted blood ultimately goes to:**

a) **The lungs ✓** b) The kidneys

c) The brain d) The liver

23. **A congenital defect with a reversed (right-to-left) shunt is called:**

a) Turner's syndrome b) Tetralogy of Fallot

c) Marfan syndrome d) **Eisenmenger's syndrome ✓**

24. **An excessive left-to-right shunt causes pulmonary vasoconstriction leading to:**

a) **Pulmonary hypertension ✓** b) Systemic hypotension

c) Mitral regurgitation d) Aortic stenosis

25. **The investigation that shows the type and size of the defect and the shunt ratio is:**

a) ECG b) **Doppler echocardiography ✓**

c) Blood culture d) Chest X-ray

26. **Irradiation and drugs during pregnancy can cause:**

a) Infective endocarditis b) Coronary artery disease

c) **Congenital heart disease ✓** d) Rheumatic fever

27. **Small septal defects can be:**

a) Treated by a pacemaker b) **Left to close spontaneously ✓**

c) Treated by CABG d) Treated by valve replacement

28. **Failure of a child with CHD to gain weight and grow normally is called:**

a) **Failure to thrive ✓** b) Cyanosis

c) Syncope d) Hemoptysis

03 Coronary Artery Disease

13 Q

29. **Myocardial ischemia means the heart muscle does not receive enough:**

a) Sodium b) **Oxygen ✓**

c) Calcium d) Vitamin D

30. **Reduced coronary blood flow is usually due to:**

a) Anemia only b) **Partial or complete blockage of the coronary arteries ✓**

c) Mitral stenosis d) Pulmonary embolism

31. **All of the following are risk factors for CAD EXCEPT:**

a) Smoking b) Diabetes

c) High blood cholesterol d) **Regular aerobic exercise ✓**

32. Regular aerobic exercise protects the heart by:

a) Raising blood sugar
b) Narrowing the coronaries
c) Increasing body weight
d) **Reducing blood pressure and cholesterol ✓**

33. Anginal pain radiates to all of the following EXCEPT:

a) Left shoulder and inner side of the arm
b) Jaw
c) **Right lower limb ✓**
d) Neck

34. The character of anginal pain is:

a) Throbbing headache-like
b) Stabbing and colicky
c) Itching
d) **Compressing, burning or squeezing ✓**

35. Anginal pain may be precipitated by:

a) **Eating a heavy meal ✓**
b) Sitting quietly
c) Drinking water
d) Sleeping

36. Atypical symptoms of ischemia are more common in:

a) Young athletes
b) Pregnant women only
c) Children
d) **Women, older people and diabetics ✓**

37. Complications of CAD include all EXCEPT:

a) Arrhythmia
b) **Pneumothorax ✓**
c) Heart failure
d) Heart attack

38. Medications used in CAD include:

a) **Aspirin, lipid-lowering drugs and nitrates ✓**
b) Antibiotics only
c) Insulin only
d) Diuretics only

39. PCI stands for:

a) **Percutaneous coronary intervention ✓**
b) Primary cardiac infarction
c) Pulmonary capillary index
d) Peripheral circulation index

40. CABG stands for:

a) Coronary angioplasty by balloon
b) Central aortic bypass graft
c) **Coronary artery bypass graft ✓**
d) Cardiac arterial balloon graft

41. Prevention of myocardial ischemia depends mainly on:

a) Bed rest
b) Avoiding all exercise
c) Antibiotics
d) **A heart-healthy lifestyle ✓**

04 Valvular Heart Diseases

45 Q

42. The number of cardiac valves is:

a) 5
b) **4 ✓**
c) 2
d) 3

43. The mitral valve allows blood flow from the left atrium to the left ventricle during:

a) Systole
b) Both systole and diastole
c) Isovolumetric contraction
d) **Diastole ✓**

44. Mitral facies appear as:

a) Yellow sclera
b) **Pinkish purple patches on the cheeks ✓**
c) Pale lips
d) Butterfly rash on the nose

45. Hoarseness of voice in mitral stenosis is due to compression of the:

a) **Left recurrent laryngeal nerve ✓**
b) Sciatic nerve
c) Facial nerve
d) Phrenic nerve

46. Systemic embolization in mitral stenosis usually occurs in patients with:

a) Hypertension
b) Sinus rhythm
c) Bradycardia
d) **Atrial fibrillation ✓**

47. Heart rate in atrial fibrillation with mitral stenosis can be controlled by:

a) Antibiotics
b) **Digoxin ✓**
c) Insulin
d) Vitamin C

48. Palpable tapping S1 at the apex is a sign of:

a) **Mitral stenosis ✓**
b) Aortic stenosis
c) Tricuspid regurgitation
d) Aortic regurgitation

49. Anticoagulation in mitral stenosis is given to prevent:

a) Hoarseness
b) Rheumatic fever
c) Hypertension
d) **Thromboembolism ✓**

50. The old surgical method for treating mitral stenosis is:

a) Pacemaker insertion
b) Heart transplantation
c) **Surgical valvotomy ✓**
d) CABG

51. Exertional dyspnea in mitral stenosis may be accompanied by:

a) Skin rash
b) Diarrhea
c) Joint swelling
d) **Cough and wheezing ✓**

52. Hemoptysis in mitral stenosis means:

a) **Coughing of blood ✓**
b) Nose bleeding
c) Vomiting of blood
d) Blood in urine

53. The interventional treatment of mitral stenosis is:

a) Pacemaker
b) TAVI
c) CABG
d) **Balloon dilatation ✓**

54. The mitral apparatus includes all of the following EXCEPT:

a) Papillary muscles
b) Chordae tendineae
c) **Aortic annulus ✓**
d) Mitral leaflets

55. In early mitral regurgitation (mild to moderate), the patient is usually:

a) Unconscious
b) **Asymptomatic ✓**
c) In shock
d) Cyanosed

56. Infective endocarditis damages the heart valve by:

a) Congenital fusion
b) Septal defects
c) Ischemia
d) **Vegetations ✓**

57. Mitral regurgitation may result from dysfunction of the:

a) **Papillary muscles ✓**
b) Coronary sinus
c) Interatrial septum
d) Pericardium

58. Mitral valve prolapse and flail leaflets are …… causes of mitral regurgitation:

a) Infective
b) **Degenerative ✓**
c) Congenital
d) Surgical

59. A mid-systolic click may occur with:

a) **Mitral valve prolapse ✓**
b) Aortic regurgitation
c) Mitral stenosis
d) Tricuspid stenosis

60. The murmur of mitral regurgitation is increased in the:

a) Standing position
b) **Left lateral position ✓**
c) Prone position
d) Sitting with legs crossed

61. The most common congenital cause of aortic stenosis is:

a) Calcification
b) Rheumatic fever
c) Syphilis
d) **Bicuspid aortic valve ✓**

62. Chest pain in aortic stenosis is due to:

a) Pleural effusion
b) Esophagitis
c) Rib fracture
d) **Imbalance between myocardial O_2 demand and supply ✓**

63. Syncope in aortic stenosis is due to:

a) High cardiac output
b) Hypertension
c) Pulmonary embolism
d) **Low cardiac output ✓**

64. Pulsus parvus et tardus means a pulse that is:

a) Large and collapsing
b) Absent
c) Irregularly irregular
d) **Small in volume and slow rising ✓**

65. Severe symptomatic aortic stenosis is treated by:

a) Diuretics only
b) Bed rest
c) Antibiotics only
d) **Valve replacement ✓**

66. The murmur of aortic stenosis radiates to the:

a) Abdomen
b) Axilla
c) **Carotids ✓**
d) Back

67. In aortic regurgitation, the apex is shifted:

a) **Laterally and downwards ✓**
b) Medially and upwards
c) Not shifted at all
d) To the right side

68. TAVI is used to treat:

a) Coronary artery disease
b) Mitral regurgitation
c) Atrial septal defect
d) **Aortic stenosis ✓**

69. Aortic regurgitation produces:

a) **Wide pulse pressure ✓**
b) High diastolic blood pressure
c) Low systolic blood pressure
d) Narrow pulse pressure

70. Patients with aortic regurgitation may feel uncomfortable awareness of:

a) **Strong heartbeats ✓**
b) Joint pain
c) Headache
d) Leg cramps

71. In aortic regurgitation, the diastolic blood pressure is:

a) **Low ✓**
b) Zero
c) High
d) Unchanged

72. Aortic regurgitation causes a characteristic pulse called:

a) **Water hammer pulse ✓**
b) Weak thready pulse
c) Pulsus parvus et tardus
d) Absent pulse

73. Carcinoid syndrome may cause stenosis or regurgitation of the:

a) Aortic valve
b) Coronary arteries
c) **Tricuspid valve ✓**
d) Mitral valve

74. Abdominal discomfort in tricuspid stenosis is due to:

a) Appendicitis
b) Gastric ulcer
c) Kidney stones
d) **Hepatomegaly ✓**

75. The murmur of aortic regurgitation is increased by:

a) Holding breath in expiration only
b) **Leaning forward ✓**
c) Lying on the right side
d) Standing

76. Tricuspid stenosis raises the pressure in the:

a) Left ventricle
b) Aorta
c) **Right atrium ✓**
d) Pulmonary veins

77. In tricuspid regurgitation, the neck veins are:

a) Calcified
b) **Congested ✓**
c) Absent
d) Collapsed

78. Tricuspid stenosis leads to:

a) **Systemic venous congestion ✓**
b) Aortic dilatation
c) Left ventricular failure
d) Pulmonary edema

79. The fluttering sensation in the neck in tricuspid stenosis is due to:

a) Goiter
b) Carotid pulsation
c) **Tall 'a' wave of JVP ✓**
d) Hoarseness

80. Tricuspid murmurs are characteristically increased with:

a) **Inspiration ✓**
b) Valsalva only
c) Expiration
d) Leaning forward

81. The murmur of tricuspid regurgitation is:

a) **Holosystolic ✓**
b) Pre-systolic only
c) Mid-diastolic
d) Continuous

82. Before valve replacement, coronary artery disease is assessed by:

a) **Cardiac catheterization ✓**
b) ESR
c) Spirometry
d) Blood culture

83. The laboratory tests for rheumatic fever and endocarditis in valvular disease are:

a) Blood sugar
b) **ESR, ASO and blood culture ✓**
c) Troponin only
d) Liver enzymes

84. A systolic thrill over the apex is felt in:

a) Pneumonia
b) Pericarditis
c) **Mitral regurgitation ✓**
d) Tricuspid stenosis

85. Medical treatment common to all valve lesions includes:

a) Bronchodilators
b) **Antibiotic prophylaxis ✓**
c) Steroids
d) Insulin

86. The mitral valve prevents backflow from the left ventricle to the left atrium during:

a) **Systole ✓**
b) Sleep only
c) Inspiration only
d) Diastole

05 Cardiac Rehabilitation

14 Q

87. All of the following are candidates for cardiac rehabilitation EXCEPT:

a) After CABG
b) Heart failure
c) **Acute myocarditis ✓**
d) Peripheral arterial disease

88. STEMI and NSTEMI are types of:

a) Congenital heart disease
b) **Acute coronary syndrome ✓**
c) Cardiomyopathy
d) Valvular heart disease

89. Core components of cardiac rehabilitation include prescribed exercise, risk factor control, lifestyle modification and:

a) **Improving psychological well-being ✓**
b) Smoking
c) High-salt diet
d) Prolonged immobilization

90. During phase I, the patient receives information about:

a) **His diagnosis, medications and work issues ✓**
b) Surgery techniques
c) Fetal circulation
d) Leisure center membership

91. Phase I includes:

a) **Early mobilization and adequate discharge planning ✓**
b) Long-term maintenance
c) Leisure center exercise
d) Structured exercise 3 times weekly

92. Phase II of cardiac rehabilitation focuses on:

a) Surgery
b) **Health education and resumption of physical activity ✓**
c) ICU care
d) Complete bed rest

93. Phase II may take the format of:

a) Surgery
b) Bed rest
c) ICU monitoring
d) **Telephone follow-up, home visits or education sessions ✓**

94. The phase sometimes erroneously referred to as the 'exercise phase' is:

a) **Phase III ✓**
b) Phase I
c) Phase II
d) Phase IV

95. Phase III combines exercise training with:

a) Complete bed rest
b) Surgery
c) **Ongoing education and psychosocial support ✓**
d) Hospital admission

96. Phase IV (maintenance) continues:

a) For 2 days
b) For 4–6 weeks
c) For 1 week
d) **Indefinitely ✓**

97. Acute myocardial infarction within …… is an absolute contraindication to exercise:

a) Two years
b) Two months
c) Ten years
d) **Two days ✓**

98. **Symptomatic severe aortic stenosis is a(n) …… contraindication to exercise:**

a) Relative
b) **Absolute ✓**
c) Not a
d) Temporary

99. **Smoking cessation, weight loss and healthy eating are part of:**

a) **Lifestyle modification ✓**
b) Medical contraindications
c) Surgical treatment
d) Fetal circulation

100. **Severe hypertension, electrolyte abnormalities and hypertrophic cardiomyopathy are:**

a) **Relative contraindications to exercise ✓**
b) Core components of rehabilitation
c) Phases of rehabilitation
d) Indications for surgery

✓ Answer Key

1 C	2 A	3 D	4 C	5 B	6 A	7 D	8 A	9 A	10 C
11 C	12 A	13 B	14 B	15 B	16 C	17 A	18 B	19 B	20 C
21 C	22 A	23 D	24 A	25 B	26 C	27 B	28 A	29 B	30 B
31 D	32 D	33 C	34 D	35 A	36 D	37 B	38 A	39 A	40 C
41 D	42 B	43 D	44 B	45 A	46 D	47 B	48 A	49 D	50 C
51 D	52 A	53 D	54 C	55 B	56 D	57 A	58 B	59 A	60 B
61 D	62 D	63 D	64 D	65 D	66 C	67 A	68 D	69 A	70 A
71 A	72 A	73 C	74 D	75 B	76 C	77 B	78 A	79 C	80 A
81 A	82 A	83 B	84 C	85 B	86 A	87 C	88 B	89 A	90 A
91 A	92 B	93 D	94 A	95 C	96 D	97 D	98 B	99 A	100 A